Context Free Grammars

Computation and Discrete Structures III

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1. Context-Free Grammars

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Context-Free Languages

A **context-free grammar** is defined as a set of 4 elements, $G = (\Sigma, N, S, P)$ where:

- Σ is the alphabet
- N are the non-terminals
- S is the start symbol
- P is the collection of rules of the form $A \rightarrow w$ where $A \in N$ and $w \in (N \cup \Sigma)^*$

Context-Free Languages

The following grammars are context-free:

$$S \rightarrow aB \mid bA$$

$$A \rightarrow a \mid aS \mid bAA$$

$$B \rightarrow b \mid bS \mid aBBS$$

Can have multiple non-terminals on the right

Context-Free Languages

$$S \rightarrow aBa \mid bAb$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow bB \mid b$$

On the right side, non-terminals can be in any position

Context-Free Languages

The following grammar is not context-free:

$$S \to aB \mid bA$$
$$aA \to aA \mid ab$$
$$bBb \to bbb \mid baS$$

Context-Free Languages

The following grammar is not context-free:

$$S \to aB \mid bA$$
$$aA \to aA \mid ab$$
$$bBb \to bbb \mid baS$$

Context-dependent grammar

Context-Free Languages

Indicate if the following regular grammars are context-free:

$$S \rightarrow aS \mid bA$$
$$A \rightarrow aB \mid bB \mid \epsilon$$
$$B \rightarrow aB \mid bB$$

$$S \rightarrow aA \mid bB \mid \epsilon$$
$$A \rightarrow bS$$
$$B \rightarrow aS$$

Context-Free Languages

$$S \rightarrow aS \mid bA$$
$$A \rightarrow aB \mid bB \mid \epsilon$$
$$B \rightarrow aB \mid bB$$

$$S \rightarrow aA \mid bB \mid \epsilon$$
$$A \rightarrow bS$$
$$B \rightarrow aS$$

Regular Grammars and Context-Free Grammars

Note that any regular grammar is also context-free. Both are context-free grammars that also satisfy regular grammar conditions

Context-Free Languages

There are context-free grammars that are not regular:

$$S \to aAb \mid bBa$$
$$A \to aAb \mid \epsilon$$
$$B \to bBa \mid \epsilon$$

$$S \to aBa \mid bAb$$
$$A \to aA \mid a$$
$$B \to bB \mid b$$

Context-Free Grammars that are not Regular

Both are context-free grammars that do not satisfy regular grammar conditions

Context-Free Languages

Regular Languages and Context-Free Languages

Every regular language is a context-free language (not vice versa)

Context-Free Languages

Regular Languages and Context-Free Languages

Every regular language is a context-free language (not vice versa)

- The language given by the expression a^*b^* is regular, therefore also context-free
- The language $\{a^n b^n \mid n \geq 0\}$ is context-free but not regular

Context-Free Languages

Consider the following context-free grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow aSb \mid ab$$

Context-Free Languages

Consider the following context-free grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow aSb \mid ab$$

The grammar generates the language $\{a^n b^n \mid n \geq 1\}$

Context-Free Languages

Consider the following context-free grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow aSa \mid bSb \mid aSb \mid bSa \mid aa \mid bb \mid ab \mid ba$$

Context-Free Languages

Consider the following context-free grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \to aSa \mid bSb \mid aSb \mid bSa \mid aa \mid bb \mid ab \mid ba$$

The grammar generates the language of all strings with an even number of symbols

Context-Free Languages

Consider the following context-free grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$$

Context-Free Languages

Consider the following context-free grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$$

The grammar generates the language of all palindromic strings over $\Sigma = \{a, b\}$

Context-Free Languages

Consider the following context-free grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow aSbS \mid bSaS \mid \epsilon$$

Context-Free Languages

Consider the following context-free grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow aSbS \mid bSaS \mid \epsilon$$

The grammar generates the language of all strings over $\Sigma = \{a, b\}$ with equal number of a's and b's

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^n c^m \mid n, m \geq 0\}$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^n c^m \mid n, m \geq 0\}$

$$S \rightarrow AC$$

$$A \rightarrow aAb \mid \epsilon$$

$$C \rightarrow cC \mid \epsilon$$

Context-Free Languages

Find a context-free grammar that generates the language
$L = \{a^n b^m \mid n < m, n \geq 0, m \geq 1\}$

Context-Free Languages

Find a context-free grammar that generates the language

$L = \{a^n b^m \mid n < m, n \geq 0, m \geq 1\}$

$$S \rightarrow aSb \mid Sb \mid b$$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^m a^n \mid n, m \geq 1\}$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^m a^n \mid n, m \geq 1\}$

$$S \rightarrow aAa$$
$$A \rightarrow aAa \mid bB$$
$$B \rightarrow bB \mid \epsilon$$

$$S \rightarrow aSa \mid aBa$$
$$B \rightarrow bB \mid b$$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^m c^m \mid n, m \geq 0\}$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^m c^m \mid n, m \geq 0\}$

$$S \rightarrow AC$$

$$A \rightarrow aA \mid \epsilon$$

$$C \rightarrow bCc \mid \epsilon$$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^n c^m d^m \mid n, m \geq 1\}$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^n c^m d^m \mid n, m \geq 1\}$

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow cBd \mid cd$$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^m c^m d^n \mid n, m \geq 1\}$

Context-Free Languages

Find a context-free grammar that generates the language $L = \{a^n b^m c^m d^n \mid n, m \geq 1\}$

$$S \rightarrow aSd \mid aAd$$

$$A \rightarrow bAc \mid bc$$

Context-Free Languages

It is not possible to find context-free grammars that represent the following languages:

- $\{a^n b^n c^n \mid n \geq 1\}$
- $\{a^n b^m c^n d^m \mid n \geq 1\}$
- $\{a^n b^m c^n \mid m \geq n\}$

Context-Free Languages

Grammar handling

- Clean grammars
- Well-formed grammars
- Chomsky normal form
- Greibach normal form

Context-Free Languages

Consider the following grammar:

$$S \rightarrow bA \mid B \mid D$$
$$B \rightarrow A \mid a$$
$$A \rightarrow aA \mid bA \mid b$$
$$C \rightarrow abd$$

Context-Free Languages

Consider the following grammar:

$$S \rightarrow bA \mid B \mid D$$
$$B \rightarrow A \mid a$$
$$A \rightarrow aA \mid bA \mid b$$
$$C \rightarrow abd$$

Useless production and useless non-terminal

- $C \rightarrow abd$ is a **useless production**
- D is a **useless non-terminal**

Context-Free Languages

Given a grammar G, the goal is to clean it. There are two states in which grammars can be transformed:

- **Clean grammars**
- **Well-formed grammars**

Context-Free Languages

Given a grammar G, the goal is to clean it. There are two states in which grammars can be transformed:

- **Clean grammars**: No useless symbols, i.e., no non-terminating or unreachable symbols
- **Well-formed grammars**

Context-Free Languages

Given a grammar G, the goal is to clean it. There are two states in which grammars can be transformed:

- **Clean grammars**
- **Well-formed grammars**: No useless symbols and also no nullable or renaming symbols

Contenido

1. Context-Free Grammars

2. Clean Grammars

Context-Free Languages

Clean grammar

A grammar is clean if it has no useless symbols

A symbol A is **useless** if it is not terminating or not reachable

- A symbol A is **terminating** if there exists $w \in \Sigma^*$ such that $A \xrightarrow{*} w$
- A symbol A is **reachable** if $S \xrightarrow{*} uAv, u, v \in (N \cup \Sigma)^*$

Context-Free Languages

Clean grammar

A grammar is clean if it has no useless symbols

A symbol A is **useless** if it is not terminating or not reachable

- A symbol A is **terminating** if there exists $w \in \Sigma^*$ such that $A \xrightarrow{*} w$
- A symbol A is **reachable** if $S \xrightarrow{*} uAv, u, v \in (N \cup \Sigma)^*$

$$S \to AB$$
$$A \to aA \mid a$$
$$B \to bB \mid b \mid bC$$
$$C \to cC \mid dC$$
$$D \to abc$$

Context-Free Languages

$$S \rightarrow Aa \mid B$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow b \mid bC$$
$$C \rightarrow eE$$
$$D \rightarrow dD \mid \epsilon$$
$$E \rightarrow cC$$

- Which symbols are terminating?
- Which symbols are reachable?

Context-Free Languages

$$S \rightarrow Aa \mid B$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow b \mid bC$$
$$C \rightarrow eE$$
$$D \rightarrow dD \mid \epsilon$$
$$E \rightarrow cC$$

- C and E are **not terminating**, i.e., cannot generate terminal symbols
- D **is not reachable**, i.e., cannot reach D from the start symbol S

Context-Free Languages

There are two algorithms to eliminate useless symbols:

- **Algorithm 1**: finds terminating symbols
- **Algorithm 2**: finds reachable symbols

Context-Free Languages

$$S \rightarrow aS \mid bB$$
$$B \rightarrow bS \mid b$$
$$C \rightarrow aB \mid bD$$
$$D \rightarrow aD \mid cD$$

Context-Free Languages

$$S \rightarrow aS \mid bB$$
$$B \rightarrow bS \mid b$$
$$C \rightarrow aB \mid bD$$
$$D \rightarrow aD \mid cD$$
$$TERM = \{B, S, C\}$$

Context-Free Languages

$$S \rightarrow aS \mid bB$$
$$B \rightarrow bS \mid b$$
$$C \rightarrow aB \mid bD$$
$$D \rightarrow aD \mid cD$$

$TERM = \{B, S, C\}$

Remove D from the grammar

Context-Free Languages

$$S \to aS \mid bB$$
$$B \to bS \mid b$$
$$C \to aB \mid bD$$
$$D \to aD \mid cD$$

$TERM = \{B, S, C\}$

Remove D from the grammar

$$S \to aS \mid bB$$
$$B \to bS \mid b$$
$$C \to aB$$

Context-Free Languages

Find the terminating symbols for the following grammar:

$$S \rightarrow Da \mid Eba \mid \epsilon$$

$$B \rightarrow bCc$$

$$C \rightarrow C$$

$$D \rightarrow bS$$

$$E \rightarrow bE$$

Context-Free Languages

Find the terminating symbols for the following grammar:

$$S \rightarrow Da \mid Eba \mid \epsilon$$

$$B \rightarrow bCc$$

$$C \rightarrow C$$

$$D \rightarrow bS$$

$$E \rightarrow bE$$

$TERM = \{S\}$
$TERM = \{S, D\}$
Therefore $\{B, C, E\}$ are not terminating and can be eliminated because terminal symbols cannot be reached from these symbols

Context-Free Languages

$$S \rightarrow Da \mid Eba \mid \epsilon$$
$$B \rightarrow bCc$$
$$C \rightarrow C$$
$$D \rightarrow bS$$
$$E \rightarrow bE$$

$TERM = \{S, D\}$
$\{B, C, E\}$ are not terminating and can be eliminated

$$S \rightarrow Da \mid \epsilon$$
$$D \rightarrow bS$$

Context-Free Languages

Find the terminating symbols for the following grammar:

$$S \rightarrow ACD \mid bBd \mid ab$$

$$A \rightarrow aB \mid aA \mid C$$

$$B \rightarrow aDS \mid aB$$

$$C \rightarrow aCS \mid CB \mid CC$$

$$D \rightarrow bD \mid ba$$

$$E \rightarrow AB \mid aDb$$

Context-Free Languages

Find the terminating symbols for the following grammar:

$$S \rightarrow ACD \mid bBd \mid ab$$
$$A \rightarrow aB \mid aA \mid C$$
$$B \rightarrow aDS \mid aB$$
$$C \rightarrow aCS \mid CB \mid CC$$
$$D \rightarrow bD \mid ba$$
$$E \rightarrow AB \mid aDb$$

$TERM = \{S, D\}$
$TERM = \{S, D, B, E\}$
$TERM = \{S, D, B, E, A\}$
C is not terminating, can be eliminated

Context-Free Languages

$$S \to ACD \mid bBd \mid ab$$
$$A \to aB \mid aA \mid C$$
$$B \to aDS \mid aB$$
$$C \to aCS \mid CB \mid CC$$
$$D \to bD \mid ba$$
$$E \to AB \mid aDb$$

$TERM = \{S, D, B, E, A\}$

C is not terminating, can be eliminated

$$S \to bBd \mid ab$$
$$A \to aB \mid aA$$
$$B \to aDS \mid aB$$
$$D \to bD \mid ba$$
$$E \to AB \mid aDb$$

Context-Free Languages

Find the terminating symbols for the following grammar:

$$S \rightarrow aS \mid aA \mid D$$

$$A \rightarrow aAa \mid aAD \mid \epsilon$$

$$B \rightarrow aB \mid BC$$

$$C \rightarrow aBb \mid CC \mid \epsilon$$

$$D \rightarrow aB \mid bA \mid aa \mid A$$

Context-Free Languages

Find the terminating symbols for the following grammar:

$$S \rightarrow aS \mid aA \mid D$$

$$A \rightarrow aAa \mid aAD \mid \epsilon$$

$$B \rightarrow aB \mid BC$$

$$C \rightarrow aBb \mid CC \mid \epsilon$$

$$D \rightarrow aB \mid bA \mid aa \mid A$$

$TERM = \{A, C, D\}$

$TERM = \{A, C, D, S\}$

B is not terminating, can be eliminated

Context-Free Languages

$$S \to aS \mid aA \mid D$$
$$A \to aAa \mid aAD \mid \epsilon$$
$$B \to aB \mid BC$$
$$C \to aBb \mid CC \mid \epsilon$$
$$D \to aB \mid bA \mid aa \mid A$$

$TERM = \{A, C, D, S\}$

B is not terminating, can be eliminated

$$S \to aS \mid aA \mid D$$
$$A \to aAa \mid aAD \mid \epsilon$$
$$C \to CC \mid \epsilon$$
$$D \to bA \mid aa \mid A$$

Context-Free Languages

Find the terminating symbols for the following grammar:

$$S \rightarrow a \mid aA \mid B \mid C$$

$$A \rightarrow aB \mid \epsilon$$

$$B \rightarrow Aa$$

$$C \rightarrow bCD$$

$$D \rightarrow ccc$$

Context-Free Languages

Find the terminating symbols for the following grammar:

$$S \rightarrow a \mid aA \mid B \mid C$$

$$A \rightarrow aB \mid \epsilon$$

$$B \rightarrow Aa$$

$$C \rightarrow bCD$$

$$D \rightarrow ccc$$

$TERM = \{S, A, D\}$

$TERM = \{S, A, D, B\}$

C is not terminating, can be eliminated

Context-Free Languages

$$S \rightarrow a \mid aA \mid B \mid C$$
$$A \rightarrow aB \mid \epsilon$$
$$B \rightarrow Aa$$
$$C \rightarrow bCD$$
$$D \rightarrow ccc$$

$TERM = \{S, A, D, B\}$

C is not terminating, can be eliminated

$$S \rightarrow a \mid aA \mid B$$
$$A \rightarrow aB \mid \epsilon$$
$$B \rightarrow Aa$$
$$D \rightarrow ccc$$

Context-Free Languages

There are two algorithms:

- **Algorithm 1**: finds terminating symbols
- **Algorithm 2**: finds reachable symbols

Context-Free Languages

$$S \rightarrow aS \mid bB$$
$$B \rightarrow bS \mid b \mid bC$$
$$C \rightarrow aB \mid a$$
$$D \rightarrow aD \mid c$$

Context-Free Languages

$$S \to aS \mid bB$$
$$B \to bS \mid b \mid bC$$
$$C \to aB \mid a$$
$$D \to aD \mid c$$

$REACH = \{S, B, C\}$

D is not reachable, can be eliminated

Context-Free Languages

$$S \to aS \mid bB$$
$$B \to bS \mid b \mid bC$$
$$C \to aB \mid a$$
$$D \to aD \mid c$$

$REACH = \{S, B, C\}$

D is not reachable, can be eliminated

$$S \to aS \mid bB$$
$$B \to bS \mid b \mid bC$$
$$C \to aB \mid a$$

Context-Free Languages

Find the reachable symbols for the following grammar:

$$S \rightarrow aS \mid AaB \mid ACS$$

$$A \rightarrow aS \mid AaB \mid AC$$

$$B \rightarrow bB \mid DB \mid BB$$

$$C \rightarrow aDa \mid ABD \mid ab$$

$$D \rightarrow aD \mid DD \mid ab$$

$$E \rightarrow FF \mid aa$$

$$F \rightarrow aE \mid EF$$

Context-Free Languages

Find the reachable symbols for the following grammar:

$$S \to aS \mid AaB \mid ACS$$
$$A \to aS \mid AaB \mid AC$$
$$B \to bB \mid DB \mid BB$$
$$C \to aDa \mid ABD \mid ab$$
$$D \to aD \mid DD \mid ab$$
$$E \to FF \mid aa$$
$$F \to aE \mid EF$$

$REACH = \{S\}$

$REACH = \{S, A, B, C\}$

$REACH = \{S, A, B, C, D\}$

Therefore $\{E, F\}$ are not reachable and can be eliminated because these symbols cannot be reached from S

Context-Free Languages

$$S \to aS \mid AaB \mid ACS$$
$$A \to aS \mid AaB \mid AC$$
$$B \to bB \mid DB \mid BB$$
$$C \to aDa \mid ABD \mid ab$$
$$D \to aD \mid DD \mid ab$$
$$E \to FF \mid aa$$
$$F \to aE \mid EF$$

$REACH = \{S, A, B, C, D\}$
$\{E, F\}$ are not reachable and can be eliminated

$$S \to aS \mid AaB \mid ACS$$
$$A \to aS \mid AaB \mid AC$$
$$B \to bB \mid DB \mid BB$$
$$C \to aDa \mid ABD \mid ab$$
$$D \to aD \mid DD \mid ab$$

Context-Free Languages

Find the reachable symbols for the following grammar:

$$S \rightarrow bB \mid aA$$

$$A \rightarrow aS \mid abC$$

$$B \rightarrow bB \mid aE \mid aa$$

$$C \rightarrow aS \mid AB$$

$$D \rightarrow aD \mid ab$$

$$E \rightarrow aA \mid aa$$

Context-Free Languages

Find the reachable symbols for the following grammar:

$$S \rightarrow bB \mid aA$$
$$A \rightarrow aS \mid abC$$
$$B \rightarrow bB \mid aE \mid aa$$
$$C \rightarrow aS \mid AB$$
$$D \rightarrow aD \mid ab$$
$$E \rightarrow aA \mid aa$$

$REACH = \{S\}$

$REACH = \{S, A, B\}$

$REACH = \{S, A, B, C, E\}$

Therefore $\{D\}$ is not reachable and can be eliminated

Context-Free Languages

$$S \to bB \mid aA$$
$$A \to aS \mid abC$$
$$B \to bB \mid aE \mid aa$$
$$C \to aS \mid AB$$
$$D \to aD \mid ab$$
$$E \to aA \mid aa$$

$$S \to bB \mid aA$$
$$A \to aS \mid abC$$
$$B \to bB \mid aE \mid aa$$
$$C \to aS \mid AB$$
$$E \to aA \mid aa$$

$REACH = \{S, A, B, C, E\}$

Therefore $\{D\}$ is not reachable and can be eliminated

Context-Free Languages

Find the reachable symbols for the following grammar:

$$S \rightarrow aB$$

$$A \rightarrow bcCCC \mid dA$$

$$B \rightarrow e$$

$$C \rightarrow fA$$

$$D \rightarrow Dgh$$

Context-Free Languages

Find the reachable symbols for the following grammar:

$$S \rightarrow aB$$
$$A \rightarrow bcCCC \mid dA$$
$$B \rightarrow e$$
$$C \rightarrow fA$$
$$D \rightarrow Dgh$$

$REACH = \{S\}$

$REACH = \{S, B\}$

Therefore $\{A, C, D\}$ are not reachable and can be eliminated

Context-Free Languages

$$S \rightarrow aB$$
$$A \rightarrow bcCCC \mid dA$$
$$B \rightarrow e$$
$$C \rightarrow fA$$
$$D \rightarrow Dgh$$

$REACH = \{S, B\}$
$\{A, C, D\}$ are not reachable and can be eliminated

$$S \rightarrow aB$$
$$B \rightarrow e$$

Context-Free Languages

Given a grammar G, the goal is to clean it. There are two states in which grammars can be transformed:

- **Clean grammars**: No useless symbols, i.e., no non-terminating or unreachable symbols
- **Well-formed grammars**

Context-Free Languages

Eliminate non-terminating symbols first, then unreachable symbols for the following grammar:

$$S \rightarrow a \mid AB$$

$$A \rightarrow aA \mid \epsilon$$

Context-Free Languages

Eliminate non-terminating symbols first, then unreachable symbols for the following grammar:

$$S \rightarrow a \mid AB$$

$$A \rightarrow aA \mid \epsilon$$

$TERM = \{S, A\}$

Therefore B can be eliminated, resulting in:

$$S \rightarrow a$$

$$A \rightarrow aA \mid \epsilon$$

$REACH = \{S\}$

Therefore A can be eliminated, resulting in:

$$S \rightarrow a$$

Context-Free Languages

Eliminate unreachable symbols first, then non-terminating symbols for the following grammar:

$$S \rightarrow a \mid AB$$

$$A \rightarrow aA \mid \epsilon$$

Context-Free Languages

Eliminate unreachable symbols first, then non-terminating symbols for the following grammar:

$$S \rightarrow a \mid AB$$

$$A \rightarrow aA \mid \epsilon$$

$REACH = \{S, A, B\}$
Therefore no symbols can be eliminated
$TERM = \{S, A\}$
Therefore B can be eliminated, resulting in:

$$S \rightarrow a$$

$$A \rightarrow aA \mid \epsilon$$

A **is not reachable**

Context-Free Languages

Elimination of useless symbols in a grammar

For a grammar G to have no **useless symbols**, non-terminating symbols must be eliminated first, then unreachable symbols

Context-Free Languages

Eliminate useless symbols for the following grammar:

$$S \rightarrow SBS \mid BC \mid Bb$$

$$A \rightarrow AA \mid aA$$

$$B \rightarrow aBCa \mid b$$

$$C \rightarrow aC \mid ACC \mid abb$$

$$D \rightarrow aAB \mid ab$$

$$E \rightarrow aS \mid bAA$$

$$F \rightarrow aDb \mid aF$$

Context-Free Languages

First eliminate non-terminating symbols:

$$S \to SBS \mid BC \mid Bb$$
$$A \to AA \mid aA$$
$$B \to aBCa \mid b$$
$$C \to aC \mid ACC \mid abb$$
$$D \to aAB \mid ab$$
$$E \to aS \mid bAA$$
$$F \to aDb \mid aF$$

$TERM = \{B, C, D\}$
$TERM = \{B, C, D, S, F\}$
$TERM = \{B, C, D, S, F, E\}$
Therefore $\{A\}$ can be eliminated

Context-Free Languages

$$S \to SBS \mid BC \mid Bb$$
$$A \to AA \mid aA$$
$$B \to aBCa \mid b$$
$$C \to aC \mid ACC \mid abb$$
$$D \to aAB \mid ab$$
$$E \to aS \mid bAA$$
$$F \to aDb \mid aF$$

$$S \to SBS \mid BC \mid Bb$$
$$B \to aBCa \mid b$$
$$C \to aC \mid abb$$
$$D \to ab$$
$$E \to aS$$
$$F \to aDb \mid aF$$

$TERM = \{B, C, D, S, F, E\}$

Therefore $\{A\}$ can be eliminated

Now eliminate unreachable symbols

Context-Free Languages

$S \to SBS \mid BC \mid Bb$

$B \to aBCa \mid b$

$C \to aC \mid abb$

$D \to ab$

$E \to aS$

$F \to aDb \mid aF$

$REACH = \{S\}$

$REACH = \{S, B, C\}$

Therefore $\{D, E, F\}$ can be eliminated

Context-Free Languages

$$S \to SBS \mid BC \mid Bb$$
$$B \to aBCa \mid b$$
$$C \to aC \mid abb$$
$$D \to ab$$
$$E \to aS$$
$$F \to aDb \mid aF$$

$REACH = \{S, B, C\}$

Therefore $\{D, E, F\}$ can be eliminated

Grammar without useless symbols

$$S \to SBS \mid BC \mid Bb$$
$$B \to aBCa \mid b$$
$$C \to aC \mid abb$$

Context-Free Languages

Eliminate useless symbols for the following grammar:

$$S \rightarrow aAb \mid cEB \mid CE$$

$$A \rightarrow dBE \mid eeC$$

$$B \rightarrow ff \mid D$$

$$C \rightarrow gFB \mid ae$$

$$D \rightarrow h$$

Context-Free Languages

First eliminate non-terminating symbols:

$$S \to aAb \mid cEB \mid CE$$
$$A \to dBE \mid eeC$$
$$B \to ff \mid D$$
$$C \to gFB \mid ae$$
$$D \to h$$

$$TERM = \{B, C, D\}$$
$$TERM = \{B, C, D, A\}$$
$$TERM = \{B, C, D, A, S\}$$

Therefore $\{E, F\}$ can be eliminated

Context-Free Languages

Now eliminate unreachable symbols:

$$S \rightarrow aAb$$

$$A \rightarrow eeC$$

$$B \rightarrow ff \mid D$$

$$C \rightarrow ae$$

$$D \rightarrow h$$

Context-Free Languages

Now eliminate unreachable symbols:

$$S \to aAb$$
$$A \to eeC$$
$$B \to ff \mid D$$
$$C \to ae$$
$$D \to h$$

$REACH = \{S\}$
$REACH = \{S, A\}$
$REACH = \{S, A, C\}$
Therefore $\{B, D\}$ can be eliminated

Context-Free Languages

The grammar without useless symbols is:

$$S \rightarrow aAb$$

$$A \rightarrow eeC$$

$$C \rightarrow ae$$

Context-Free Languages

Eliminate useless symbols for the following grammar:

$$S \rightarrow A \mid AA \mid AAA$$

$$A \rightarrow ABa \mid ACa \mid a$$

$$B \rightarrow ABa \mid Ab \mid \epsilon$$

$$C \rightarrow Cab \mid CC$$

$$D \rightarrow CD \mid Cd \mid Cea$$

$$E \rightarrow b$$

Context-Free Languages

Eliminate useless symbols for the following grammar:

$$S \rightarrow A \mid AA \mid AAA$$
$$A \rightarrow ABa \mid ACa \mid a$$
$$B \rightarrow ABa \mid Ab \mid \epsilon$$
$$C \rightarrow Cab \mid CC$$
$$D \rightarrow CD \mid Cd \mid Cea$$
$$E \rightarrow b$$

$TERM = \{A, B, E\}$

$TERM = \{A, B, E, S\}$

Therefore $\{C, D\}$ can be eliminated

Context-Free Languages

Now eliminate unreachable symbols:

$$S \rightarrow A \mid AA \mid AAA$$

$$A \rightarrow ABa \mid a$$

$$B \rightarrow ABa \mid Ab \mid \epsilon$$

$$E \rightarrow b$$

Context-Free Languages

Now eliminate unreachable symbols:

$$S \to A \mid AA \mid AAA$$
$$A \to ABa \mid a$$
$$B \to ABa \mid Ab \mid \epsilon$$
$$E \to b$$

$$REACH = \{S\}$$
$$REACH = \{S, A\}$$
$$REACH = \{S, A, B\}$$

Therefore $\{E\}$ can be eliminated

Context-Free Languages

The grammar without useless symbols is:

$$S \rightarrow A \mid AA \mid AAA$$

$$A \rightarrow ABa \mid a$$

$$B \rightarrow ABa \mid Ab \mid \epsilon$$

References

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